Beneath the Heavens: A Thorough Measurement Study of the Starlink Terrestrial Network

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2025 IEEE/ACM International Symposium on Quality of Service (IWQoS) | 979-8-3315-4940-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/IWQOS65803.2025.11143359

***Abstract*—The emerging low earth orbit (LEO) satellite Internet has gained worldwide popularity. Starlink, a prominent LEO satellite Internet service provider, has attracted the most users due to its low latency, wide coverage, and strong usability. Current research on Starlink primarily focuses on the space segment and its impact on network performance. However, as an essential component, the architecture and unique features of the Starlink Terrestrial Network (SLTN) are not well-explored, which significantly affects the performance, security, and development of the whole network. In this paper, we fill this gap by conducting a thorough measurement study to profile the SLTN from various aspects and reveal its potential effects, with specific attention to the network assets, topology, and routing strategies.**

We developed a novel framework including active and passive measurement methods for collecting multiple network assets, tracing different route paths, and scanning active service of the SLTN. The open source intelligence was utilized for the first time to collect extensive real-user network status. Leveraging these techniques, we observed a rapid expansion of the Starlink service with the latest network assets, collected in January 2025, including over 23.5K /28 IP prefixes residing in 143 countries. We mapped the consistent topology of the global Starlink Internet access service and identified the specific IP addresses associated with the four types of network routing nodes. Multiple internal routing strategies were uncovered, which facilitate direct user-to-user interactions. Particularly, we revealed the switches of terrestrial infrastructure that users connected and the changes in routing strategies, which should be considered in future quality of service (QoS) evaluations. Our measurement also provides methods for improving users' perceptions and serves as a basis for studies like security risk evaluation and service discovery.

***Index Terms*—Satellite-Terrestrial Integrated Network, Starlink, Internet Measurement**

I. INTRODUCTION

Satellite Internet provides Internet access service by utilizing communication satellites, offering broadband access with extensive coverage. Compared with geostationary satellites, LEO satellites are positioned closer to the surface, which reduces network latency and improves service quality [1]. Starlink, operated by SpaceX, has been publicly available since February 2021, establishing a vast LEO satellite network that serves over 100 countries and areas. By September 2024, Starlink has deployed over 7K LEO satellites and reached more than 4 million subscribers, making it the leading satellite Internet service provider in terms of coverage, latency, and user-friendliness. Moreover, Starlink has been utilized for military communications during the Russo-Ukrainian War [2].

The Starlink network's infrastructure consists of space and terrestrial segments. The space segment is primarily composed of a series of low-orbit satellites, while the terrestrial segment encompasses the User Terminal, Ground Station (GS), and Internet Point of Presence (PoP). Studies of the Starlink network primarily focus on the space segment. A large group of researchers measured the performance of the Starlink network from various aspects [13] [15] [16] and explored how spatial factors, like weather and satellite coverage, affect the quality of service (QoS) [11] [17]. Novel network protocols have been designed to align with the performance requirements [21] [22]. The terrestrial segment, as an essential component, is also crucial in data transmission (Fig. 1). Its special network architecture could greatly affect the QoS evaluation, security analysis, and development of the whole Starlink network ecosystem. However, to the best of our knowledge, there is a significant lack of understanding of the terrestrial network and the effects. Furthermore, insufficient comprehension has already misled some applied research, such as the scanning of irrelevant hosts in security risk evaluations [9] [10].

To fill this gap, we conducted a thorough measurement study of the Starlink Terrestrial Network. Novel methods were proposed to collect sufficient research data. We then presented an in-depth analysis, with special attention to network assets, topology, and routing strategies. The effect of the SLTN on QoS evaluation, security, etc., is given significant attention.

Profiling the SLTN is by no means trivial since it is a black-box system with a wide distribution and network fluctuations. To this end, we proposed a specific measurement framework with active and passive measurement methods to collect effective research data. We applied **O**pen **S**ource **Int**elligence (OSINT) to collect extensive real-user network status data for the first time, addressing the challenge of limited volunteers while broadening measurement coverage. The active measurement methods consist of *StarCollect*, a method for collecting comprehensive network assets continuously and automatically; *StarTrace*, a specialized network path tracing method for discovering multiple route paths of the SLTN; and *StarScan*, an active host scanning tool that efficiently identified open services of important network routing nodes.

Leveraging the collected data, we profiled the SLTN from various aspects, along with a set of insightful findings affecting the QoS evaluation, security evaluations, etc. During our measurement between January 2023 and January 2025, we observed a rapid expansion of the Starlink service according to associated network assets. The latest data includes over 23.5K /28 IP prefixes (a 48% growth), 290 domain names, and 6 autonomous systems (AS), residing in 143 countries and regions both supporting Starlink service and under construction. Based on the statistics of five types of route paths, we mapped the consistent topology of the global Starlink Internet access service and identified four types of network routing nodes. Besides, we proposed an efficient classification algorithm, determining that 79.6% of the IP addresses were associated with Starlink *user nodes* and 2.3% associated with the PoP *exit nodes*. Moreover, multiple internal routing strategies were identified, which facilitate direct user-to-user interactions without traditional Internet transit. More importantly, our findings help to improve users' perceptions of the current network status, such as identifying the specific terrestrial infrastructure they are connecting through the network path tracing method. The measurement results on network architecture serve as a basis for studies like security risk evaluations and service discovery. For instance, we observed there were open BGP and router management services exposed to the Internet, which bring security risks and should be protected. In particular, we revealed the switches of PoP that Starlink users connected and the changes in routing strategies of the SLTN, which should be considered in QoS measurements and evaluations.

The contributions of the paper are summarized as follows:

- *New Techniques:* We developed a specialized measurement framework for the SLTN to collect associated network assets, discover network route paths, and implement host scanning. We first utilized OSINT to collect wide real-user network status data. The framework provides novel insights for the measurement of other satellite Internet systems.
- *New Findings:* We collected sufficient research data and distilled a set of insightful findings specifically in network assets, topology, and routing strategies of the SLTN.
- *Practical Applications:* We proposed practical applications. Our study serves as a crucial basis for improving user perception and studies like security risk evaluation. The findings on routing variations also facilitate QoS evaluations.

The rest of the paper is organized as follows: §II present an overview of the Starlink network. In §III, we propose focused research questions and the corresponding measurement methodology. §IV details our in-depth analysis of the SLTN. The applications of our measurement study are shown in §V. §VI presents the ethical issues. §VII summarizes previous works related to the SLTN. §VIII concludes this paper.

II. BACKGROUND

Satellite Internet has emerged as a pivotal technology for global connectivity, offering broadband access in remote and underserved regions. Traditionally, geostationary satellites have been the primary means of satellite-based Internet, providing wide coverage but facing high latency due to their distance. In contrast, low earth orbit satellites are positioned much closer to the Earth's surface, reducing latency and improving

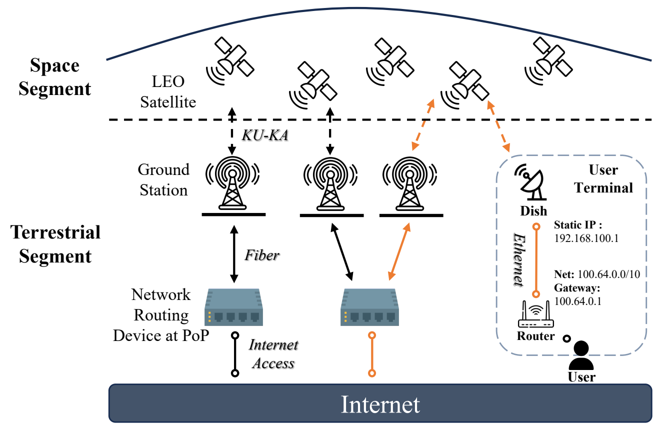


Fig. 1. **The network infrastructure supporting the Starlink service in each area. The orange lines represent the network transmission link of the Internet Access Service.**

overall service quality. Starlink, operated by SpaceX, LLC, has been available to the public since February 2021. It has established a comprehensive network system with LEO satellites, providing satellite Internet access to over 100 countries and territories. Since early 2022, the Starlink program has emerged as the preeminent Satellite Internet System regarding satellite count and overall scale. As of September 2024, SpaceX has successfully deployed over 7K LEO satellites and announced reaching more than 4 million subscribers. Starlink emerged as the preeminent satellite Internet system because of its extensive coverage, low latency, and high user-friendliness. Moreover, Starlink has been used in the Russo-Ukrainian War for military communications [1] [2]. A military version, *Starshield*, is designed for government use.

Fig. 1 illustrates the infrastructure of Starlink and the entire data transmission with network information. To enable Internet access, data is exchanged among several neighboring infrastructures [2]. In this study, the SLTN refers to the network based on the interconnections of the Starlink terrestrial segment infrastructure.

The user is required to acquire a user terminal to access the Starlink network. Data packets originating from the user's PC are transmitted through the user terminal as KU-KA band signals to a satellite. This satellite subsequently relays the packets to a nearby GS. The GS then routes these packets via wired fiber to a connected PoP, which forwards them to the Internet [3]. The details are summarized as follows.

User Terminal. The user terminal comprises a router and a dish connected via Ethernet. It converts the Ethernet packets from the user's network devices into satellite signals in the KU-KA band and transmits them to the LEO satellite. The router's WAN port acquires network addresses by parsing DHCP packets from the Starlink network. Owing to the scarcity of IPv4 resources, Starlink employs CGNAT technology. Regular users receive private IP addresses, whereas premium users can switch to dedicated public IP addresses. Following the Starlink routing algorithm, the dish converts users' Ethernet frames and sends them to a specific satellite.

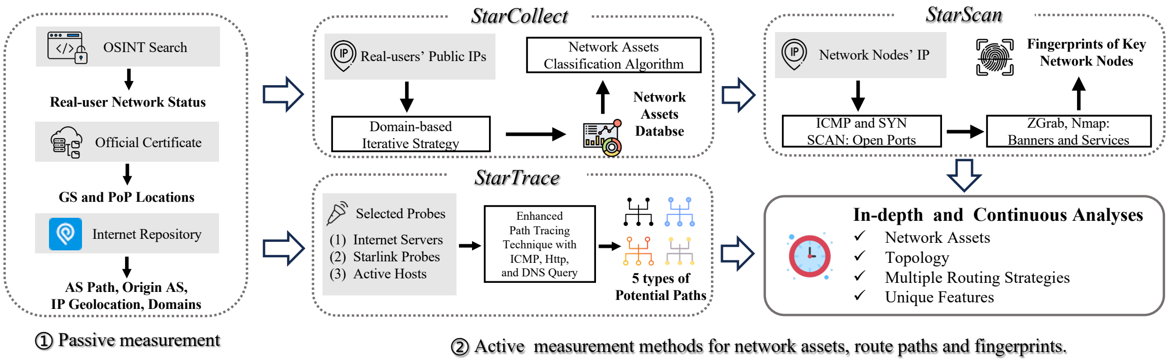


Fig. 2. Our measurement framework with active and passive measurement methods.

Satellite and Ground Station. After receiving user data, the satellite selects a specific nearby GS and transmits the data to that GS via KU-KA band signals. However, wireless signals have inherent distance limitations. For example, satellites over the US cannot directly communicate with the GS in Europe. Starlink has deployed over 150 GS worldwide to expand its service coverage. Meanwhile, Starlink is exploring reducing the number of GS with the technique of inter-satellite laser link (ISL). ISL is still under evaluation and experiment with limited scale, we do not consider its impact in this paper [2].
Internet Point of Presence. The GS employs wired fiber to transmit data packets to network routing devices at the PoP. Specifically, these routing devices bridge the internal Starlink network to the Internet

III. STUDY DESIGN AND METHODOLOGY

We first proposed focused research questions (RQ) for profiling the SLTN. To meet the needs of this measurement study, we designed a specialized framework to collect various data, with active and passive measurement methods. The data was utilized for in-depth analysis of the SLTN in §IV and will facilitate further applications and research discussed in §V.

A. Research Questions

Our research is driven by the following RQs:
RQ1: The Network Assets. As a crucial component and an identifier of a network system, we wonder what network assets are associated with the SLTN, including AS information, IP addresses, domain names, and other network identifiers. The network assets and their attributes can also reveal the scale, distribution, and key characteristics of the SLTN system.
RQ2: The Internet Access Topology. Starlink has constructed a novel Internet access network. We wonder about the characteristics of the topology, including identifying connectivity relationships among important network routing nodes and their unique roles. The effect of this type of topology on QoS is also an interesting topic.
RQ3: Network Assets Classification. We wonder whether there is an effective approach to classify a public IP address to its associated network routing node within the SLTN. The classification has important applications in fingerprinting and finding potential security risks via active network services.
RQ4: Multiple Routing Strategies. The Starlink service has an extensive distribution. We wonder whether multiple internal routing strategies are supported to facilitate direct user-to-user interactions without traditional Internet transit.

B. Challenges and Methodology Overview

Due to the characteristics of the Starlink network, we encountered several challenges in collecting research data.

- Starlink is a closed-source system with a lack of transparency in network protocols and routing algorithms, making traditional measurement methods inapplicable.
- The high deployment cost of user terminals poses challenges to the traditional measurement way of deploying widespread network probes. An efficient approach is required to collect extensive real-user data.
- The ongoing development of Starlink necessitates specific methods to continuously and effectively monitor the changes in network infrastructure and network resources.
- The movement of LEO satellites and other potential factors lead to network fluctuations, including latency and data packet reachability, which means specialized methods are required to ensure data reliability.

These factors were critical considerations throughout our methodology design. In short, as shown in Fig. 2, we proposed an automated framework *StarCollect* for collecting multidimensional network assets continuously and automatically, a specialized network path tracing method *StarTrace* for discovering multiple route paths, and an active host scanning tool *StarScan* that efficiently discovers active network services. In addition to the active measurement methods, we are the first to leverage OSINT to obtain real users' network status, mitigating the lack of volunteers while extending the measurement coverage. Other relevant datasets include GS and POP Locations, AS Path, Origin AS, IP Geolocation, and Passive DNS. A detailed explanation is provided below.

C. Active Measurement

a) StarCollect: We designed a framework, *StarCollect*, for the collection of a wide range of network assets from multiple aspects, ultimately creating a comprehensive **N**etwork **A**sset **Data**base (NADB). In other words, *StarCollect* implements a binary classification algorithm to distinguish network

TABLE I
THE NETWORK PROBES WITHIN THE SLTN.

Country	# Number	Country	# Number	Country	# Number
US	21	BE	2	MG	1
FR	11	NL	2	JP	1
CA	8	HT	2	HU	1
DE	6	CL	2	CZ	1
GB	4	KI	2	PL	1
IT	3	CO	1	GR	1
AU	3	ES	1	BJ	1

TABLE II
SELECTED PROBES AND TARGETS FOR NETWORK PATH TRACING.

Index	Potential Route Paths	Probe	Target
Path-1	Starlink user → Internet	Starlink user terminals	controlled cloud servers
Path-2	Internet → Starlink user	controlled cloud servers	Starlink user terminals
Path-3	Starlink users under the same GS	Starlink users from the same city	
Path-4	Starlink users under different GS but the same PoP	Starlink users of the same IP geolocation from nearby cities	
Path-5	Starlink users under different PoPs	Starlink users of the same different IP geolocations from different countries	

resources associated with Starlink from other networks. The database contains SLTN-related network identifiers including IP addresses, autonomous systems, domain names, IP geolocations, etc. Specifically, *StarCollect* can continuously and automatically monitor network assets' changes.

The input of *StarCollect* is a number of Starlink real-users' public IP addresses from our OSINT data. We then use an iterative algorithm to collect the whole assets. At the initial stage, we utilize the Origin AS data to map the input to their corresponding AS. The IP addresses announced by these AS are then inserted into the database. Our iterative strategy is based on the following domain-level insight: (1) We found 95% of the input IP addresses were associated with domains in the format of *customer.*.*$\{$*pop, mc*$\}$*.starlinkisp.net*, subdomains of *starlink.isp.net*. (2) The rest IP addresses are associated with *undefined.hostname.localhost*. Consequently, the next step is to retrieve the subdomains of *starlinkisp.net* and their associated IP addresses with the Passive DNS dataset. The initial stage will be repeated until no new IPs are inserted into the database, indicating the algorithm has reached convergence. We classify these IP addresses using a prefix length of 28 for ease of data deduplication. Additionally, we looked up IP geolocations as part of the database using the IP Geolocation dataset. The NADB constructed by *StarCollect* not only provides information on ASs, IP addresses, domain names, and IP geolocations but also details the relationship mapping among these different types of assets. The database proved effective through subsequent in-depth analyses in §IV-C.

Metadata of the NADB. Since 2023, we have been continuously collecting the data. We show the metadata for three key time points. In May 2023, we observed 15,804 /28 IP prefixes, 266 hostnames, and associated 107 countries and areas. In May 2024, the number is 20,828; 278; and 130. In January 2025, the number is 23,562; 290; and 143. A comprehensive analysis of the NADB is presented in §IV-A.

b) StarTrace: Considering the fluctuations of the Starlink network, we designed a specialized network path tracing method *StarTrace*. It was used to discover multiple potentially supported route paths in the Starlink network and the corresponding route-level network information.

StarTrace uses a route-tracing technique based on TTL-limited probes [26]. We employed the following improvements to address the corresponding issues: (1) In the Starlink network, the TTL response strategy adopted by network routing nodes may cause traditional techniques to fail. We complemented the responses using ICMP ECHO packets, DNS queries, and HTTP GET requests. These diverse packets can effectively provide additional route path information and bypass potential network firewalls. (2) We found that the Starlink network's high fluctuations can lead to significant variability in tracing results, specifically route paths and network latency. To enhance stability, we perform repeated path tracings. Experiments showed that over 80% of the path tracing results were consistent after repeating the tests continuously more than 15 times. This experiment result was true for different times throughout the day. Consequently, we selected the most frequent paths from the 15 identical path tracings and calculated the average network latency.

Before the start of *StarTrace*, based on the potential communication link we want to discover, we need to carefully select a probe for sending tracing packets and a target IP address. First, we gathered a set of sufficient network resources for probing: (1) *Internet Servers*: 7 cloud servers deployed in five across Asia, Europe, Oceania, South America, and North America. (2) *Active Hosts*: IP addresses of active hosts within the Starlink network identified by *StarScan*. (3) *Starlink Probes:* Ripe public probes which are voluntarily hosted by Starlink users [39] [40]. We used a total of 70 probes across over 20 countries, illustrated in Table I. Each Ripe probe also offers the public IP address and their city-level geographical location. (4) *Specific volunteers* with Starlink user terminals recruited in specific countries. Next, as illustrated in Table II, we carefully defined five distinct types of route paths that Starlink may potentially support and selected appropriate probes and detection targets for each path. *PATH-1* presents the link from Starlink users to Internet servers. Thus, we used *Starlink Probes* to visit *Internet servers*. *PATH-2* presents the link from Internet servers to Starlink users. Thus, we used *Internet servers* to visit *Active Hosts*. *PATH-3* presents the communication link of users under the same GS. Based on our analysis in §IV-A and §IV-B, we recruited Starlink volunteers from the same city with the same IP geolocation to visit each other. *PATH-4* presents the link of users under different GS but the same PoP. Correspondingly, we recruited volunteers who had the same IP geolocations but from nearby cities in the country. *PATH-5* presents the link of users under different PoPs. We used *Starlink Probes* of the different IP geolocations from different countries to visit each other.

We conducted a large-scale path tracing since early 2023, sending a total of over 50K probe packets to obtain various route paths. According to our subsequent statistical analysis in §IV-D, our selected probes and path tracing targets proved to be effective. In other words, the tracing results' statistics of the same route path are consistent, but statistics of different potential route paths are significantly different, which ensures the reliability of *StarTrace*.

c) StarScan: We developed an active host scanning tool *StarScan*. The tool efficiently scans for active hosts and extracts host features, including device banners and specific services hosted on open ports.

StarScan integrates several existing active scanning techniques. We first send ICMP and TCP SYN probe packets to scan the first 1024 ports of the target device to determine open ports. Next, Nmap [32] is used to send up to 16 different TCP, UDP, and ICMP probe packets to these open ports, aiming to elicit a response from the device and capture a potential device banner. Based on the accumulated database of fingerprints, Nmap then provides a detailed explanation of the services hosted on the device. We also used Shodan [47], which regularly scans over 1,500 ports of nearly all Internet hosts, to find additional service information.

To mitigate the impact of geographic factors on the measurement [38], we purchased 7 cloud servers in five regions across Asia, Europe, and North America, and performed scanning on the IPv4 addresses collected by *starCollect*. We found the active hosts changed rapidly, and there were over 64.9K open ports during our scanning in January 2025. Meanwhile, in early 2023, the number was only 3.7K.

D. Passive Measurement

This section explains our passive measurement, specifically collecting effective data from different open source platforms.

Open Source Intelligence. The Starlink network has an extensive terrestrial infrastructure worldwide. Undoubtedly, deploying numerous probes through volunteer recruitment is cost-prohibitive. To this end, we are the first to utilize open source intelligence to gather information on the Starlink IP address allocation and network routing. OSINT comprises information from publicly accessible sources such as government documents, news reports, social media, etc. We collected network status from Starlink real users from over 20 countries, which improves both the comprehensiveness and accuracy of our measurement. Furthermore, variations in OSINT data could indicate the developmental trajectory of the SLTN. The ethical issues will be discussed later.

We utilized the following tricks to collect effective network information. Firstly, we observed that Starlink users often test network performance using traceroute [26] to Internet hosts and share the results in the community [23] - [25]. Since the default gateway of the user router and the terrestrial gateway are officially configured as *192.168.100.1* and *100.64.0.1* respectively [5], we used these two IP addresses as keywords for searching within the community, including posting, comment, and media section. Overall, we manually collected a total of 87 users' traceroute results. Furthermore, a few Starlink users inadvertently exposed their public IP address and geographic location while addressing network issues. We conducted a keyword search to collect the data, focusing on keywords including 'DHCP', 'network troubleshot', 'public IP', etc. In addition, several Starlink users, serving as network probes, have voluntarily disclosed their public IP addresses and city-level geographic locations [39] [40]. Overall, we manually collected a total of 120 users' public IP addresses and corresponding geographic locations covering over 20 countries and regions across six continents.

Based on the OSINT data, we also inferred that the Starlink operator implemented a standardized adjustment to the SLTN network architecture at the end of 2022. During the data cleaning, we found that users' traceroute results released before December 2022 were significantly different from those afterward. The route paths released after 2023 show significant statistical consistency, discussed in §IV-B, whereas pre-2022 data displays variability. Thus, the OSINT data we used above in this study was all released in 2024 or later to reflect the latest network architecture.

GS and PoP Location. Areas with access to the Starlink service necessitate supporting ground infrastructure and network resources. The collection of GS and PoP locations helps to understand the service scale and associated network assets. Our collection primarily stems from several novel observations. Firstly, the construction of GS requires legal authorization. For example, in the US, the Federal Communications Commission is empowered to issue these access certificates [27]. Besides, for the construction of PoP, network exchange centers typically maintain records of network edge node deployments [34]. Secondly, volunteers report the infrastructure construction in Starlink user communities [23] - [25]. Additionally, social media also contribute to this dataset [28]. After de-duplicating the data, we amassed 137 GS worldwide and 27 PoP in 16 countries with city-level locations and registration details.

AS Path and Origin AS. To map an IP address to its original AS, we utilized the data published by Routeviews [29] to extract the AS paths of each IP prefix. We collected metadata from Thyme [30] to ensure data accuracy. In instances of inconsistency, we manually validated a small sample of AS paths to minimize errors through traceroute [26].

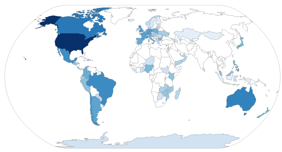
IP Geolocation and Passive DNS. IP geolocation refers to the process of determining the geographical location of an Internet-connected device based on its IP address, utilizing WHOIS records, network routing data, etc. IPinfo [31] is a trusted IP data provider. We utilized IPinfo's data to map an IP address to its IP geolocation (country, city, etc.) and find associated domain names.

IV. IN-DEPTH ANALYSIS OF THE STARLINK TERRESTRIAL NETWORK

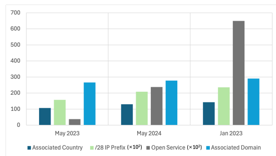
We analyzed the collected research data from various aspects to report an in-depth analysis of the SLTN and answer the RQs proposed in §III-A.

A. The Network Assets

StarCollect had classified the network assets associated with the SLTN. The latest NADB (January 2025), as shown in



(a) Distribution.



(b) Metadata.

Fig. 3. Metadata of the Starlink network assets.

Fig. 3, mainly included over 23.5K /28 IP prefixes, 290 domain names, and the city-level geographic locations of these network identifiers. These network assets are associated with 143 countries and regions, including 113 regions Starlink service has officially supported and those under construction [6]. We correlated these assets from various aspects to reveal the scale, distribution, and characteristics of the SLTN.

The SLTN is associated with several different ASs, including SpaceX AS14593 AS45700 AS27277, Google AS36492, and Trans-System INC AS14597. This discovery confirms Starlink's public statements regarding its collaboration with network operators like Google, to provide faster Internet access service [33]. AS14593 is central to the SLTN. On the one hand, it has the most significant asset scale, with an announcement of 329,984 IPs across 131 countries, demonstrating the global coverage of the Starlink service. On the other hand, §IV-C indicates that over 95% of the critical network nodes, that we identified, use IP addresses originating from AS14593. AS14593 peers with over 370 ASs, including 29 upstreams and only four downstream [33] [17], aligned with typical ISP design practices [42]. This indicates that the Starlink network, as an ISP, is designed to offer the best possible service to its customers. In addition, we detected several ASs through the identifiers (SpaceX, Starlink, etc.), including SpaceX AS27277, Starlink AS142475, and SpaceX AS397763. However, these ASs were only observed in the OSINT data up to December 2021 and are not included in our latest network asset database, possibly phased out due to the development of the Starlink network. Meanwhile, the growth in network assets highlights the ongoing evolution of the Starlink network systems. Compared with January 2023, there has been a 48% increase in the number of associated IP addresses, a 52% rise in the number of countries covered, and an 182% growth in the regions providing services.

In addition, we preliminarily explored the locations of the real Starlink users with OSINT data. We noted that there is a notable disparity, where 96% of users' IP geolocation (public IP addresses) corresponds to a geographically proximate PoP rather than their real geographic location. For instance, one Starlink user geographically located in Birmingham had an IP geolocation of Atlanta, where a PoP is situated. We attributed this disparity to the limitations of existing IP geolocation methods, most of which can only provide PoP-level accuracy [45]. Meanwhile, geographically proximate users shared the same nearby PoP, which aligns with the Starlink working principle. The other 4% of the Starlink users were from countries without Starlink infrastructure, relying on the infrastructure of neighboring countries. For instance, one user geographically located in Mongolia had an accurate IP geolocation of Mongolia, where the IP geolocation service providers may correct the data through WHOIS [46].

Answer to RQ1: The SLTN is associated with a few AS operated by SpaceX and its partners and with multiple network assets including over 23.5K /28 IP prefixes. The global AS14593 plays a crucial role, reflecting typical ISP features. The significant distribution (143 countries) and the increase in network assets indicate a rapid expansion of the Starlink service.

B. The Internet Access Topology

Using StarTrace, we got the route paths from Starlink users to Internet servers (*Path-1*) and the paths from Internet servers to Starlink users (*Path-2*). Table III illustrates the statistics of these route paths, including network routing nodes and IP prefixes. Subsequently, we analyzed these statistics to map the Starlink Internet access topology and uncovered the switch of the PoP Starlink users connected.

While different regions rely on separate infrastructure and network resources, their topological statistics are consistent. We presented the mapped Starlink Internet access network topology in Fig. 4, based on Table III. The topology indicates that *Path-1* and *Path-2* demonstrate a distinct symmetry, with adjacent network routing nodes belonging to the same network. The finding aligns with the bidirectional communication behavior. Through querying the ASN, we discovered that the data packets, transmitted from Hop 0, passed through three routing nodes within the Starlink network (AS14593). The IP address of each routing node is associated with specific IP prefixes. Hops 0 to Hop 2 belong to the internal Starlink network utilizing private IP addresses, while Hop 3 serves as the connection point between the Starlink network and the Internet. Specifically, Hop 0 (*user node*) represents the router of Starlink users, Hop 1 (*gateway node*) serves as the user's default gateway [5], and Hop 3 (*exit node*) corresponds to a network device at the PoP. We refer to Hop 2 as *link node*. The network latency from Hop 0 to Hop 1 always accounts for over 67% of the total within the internal network (Hop 0 to Hop 3), which is attributed to satellite wireless communication.

We then conducted further exploration of IP geolocation and network latency. Based on the topology, Starlink users can identify the PoP they connect to by querying the IP geolocation of the hop 4 *exit node*. Through this method, we observed there were switches of the PoP that users connected. For instance, Starlink users in Mongolia connected to the PoP in Germany before October 2024, but afterward, it switched to Japan. We double-checked the switch with network latency from our cloud server in Berlin, Germany, which increased from an average of 160.63 ms to 411.93 ms. The observation also indicated that the switch of the connected PoP can affect the network latency.

TABLE III
THE ROUTE PATHS STATISTICS OF STARLINK INTERNET ACCESS SERVICE.

HOP	0	1	2	3	4
Path-1	{User} *User Node*	100.64.0.1 *Gateway Node*	172.16.0.0/16 *Link Node*	206.224.64.0/19, 149.19.108.0/23 *Exit Node*	{Internet} /
Path-2 (reverse)	{User} *User Node*	{no response} *Gateway Node*	206.224.64.0/19, 149.19.108.0/23 *Link Node*	206.224.64.0/19, 149.19.108.0/23 *Exit Node*	{Internet} /
NOTE	/	Private IPs	Private IPs and Public IPs of AS14593	Public IPs of AS14593	Internet and other ASs

PATH-1 presents the link from Starlink users to Internet servers.
PATH-2 presents the link from Internet servers to Starlink users.

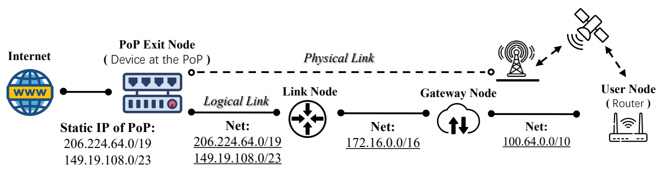


Fig. 4. The topology of the Starlink Internet access service.

Answer to RQ2: The global Starlink Internet access service has a consistent topology. It consists of four essential network routing nodes, collectively forming the link from the internal network to the Internet. Each routing node exhibits a unique consistency regarding IP addresses. Through the topology, we found there were switches of the users' connected PoP, which could affect the network latency.

C. Network Assets Classification

We here introduce an effective rule-based heuristic algorithm to classify a given public IP address to the associated SLTN network routing node. The algorithm, illustrated in Algorithm 1, helps us understand the role of each type of network routing node and facilitates security evaluation studies. We verified the accuracy of the algorithm by analyzing the network services that each type of routing node hosted.

For determining the associated IP addresses of *user nodes*, in addition to *PATH-2* path tracing, we can utilize RFC 8805 Standard [35] that Starlink follows. In other words, Starlink periodically publishes a set of public IP addresses, which is claimed to be reserved for Starlink users to improve the user experience [36]. Notably, we discovered that these released IP addresses of RFC 8805 constitute a subset of our NADB. Furthermore, the domain format, *customer.*.{pop, mc}.starlinkisp.net* and *undefined.hostname.localhost*, is another key factor. The determination of *exit nodes* is primally based on the statistics of path tracing illustrated in Table III. *StarCollect* implemented the algorithm in our NADB. The result showed that each IP address was determined uniquely to one type of network routing node. Specifically, 79.6% of the IP addresses are associated with *user nodes*, 2.3% with *exit nodes*, and the rest are not associated with any node, possibly deprecated or for other unknown uses.

Algorithm 1 Network Assets Classification

```
Input: A Public IP in the NADB
Output: The Associated Network Routing Node
Domain ← PassiveDNS(IP)
NameSet ← {*.starlinkisp.net, *.hostname.localhost}
UserSet ← Store(StarlinkRFC8805)
ExitSet ← { 206.224.64.0/19, 149.19.108.0/23 }
if IP in UserSet and Domain in NameSet then
    RoutePath ← PathTracing(IP)
    if IP matches HOP 0 in RoutePath then
        Return User Node
    end if
end if
if IP in ExitSet then
    RoutePath ← PathTracing(IP)
    if IP matches HOP 2 or HOP 3 in RoutePath then
        Return Exit Node
    end if
end if
Return Unknown Network Node
```

We confirmed the reliability of this classification algorithm through the active network services. We conducted host scanning of the IP addresses associated with the *user nodes* and *exit nodes* through *StarScan*. The results indicated that there were over 64k open ports of *user nodes*. The top services hosted in these ports are HTTP (24.5%), the services of wireless routers (7.3%), and SSH (6.5%), which are typically hosted by individual users rather than network devices. However, for *exit nodes*, only 9 open ports hosting BGP service were detected, which was used for interaction with other ASs. We inferred that network protection measures have been implemented to prevent the vital routing nodes from scanning. Moreover, we searched the Starlink router's default IP address *192.168.1.1* in the device banners. Several management interfaces of the Starlink router were found exposed to the public Internet, potentially allowing external attackers to manipulate them easily. While the risk has been revealed in previous studies [43], it remains prevalent.

Answer to RQ3: We proposed an efficient classification algorithm, determining that 79.6% of the IP addresses in the NADB were associated with *user nodes* and 2.3% with the *exit nodes*. These two types of routing nodes hosted over 64K open services, mostly related to Web and SSH. Meanwhile, the exposed BGP and router management services brought security risks.

D. The Multiple Routing Strategies

In this section, we first identified the routing strategies implemented to facilitate the internal communication of Starlink users. The analysis also revealed the Starlink internal backbone network, as observed in previous research [7]. Then, we explored the routing strategy presented by this backbone

TABLE IV
STATISTICS OF THE POTENTIAL ROUTE PATHS WE IDENTIFIED WITHIN THE SLTN.

HOP	0	1	2	3	4	5	6
Path-3	{User} *User Node*	100.64.0.1 *Gateway Node*	{Target} *User Node*				
Path-4	{User} *User Node*	100.64.0.1 *Gateway Node*	172.16.0.0/16 *Link Node*	206.224.64.0/19, 149.19.108.0/23 *Exit Node*	206.224.64.0/19, 149.19.108.0/23 *Link Node*	172.16.0.0/16 *Gateway Node*	{Target} *User Node*
Path-5	{User} *User Node*	100.64.0.1 *Gateway Node*	172.16.0.0/16 *Link Node*	206.224.64.0/19, 149.19.108.0/23 *Exit Node*	206.224.64.0/19, 149.19.108.0/23 *Exit Node*	206.224.64.0/19, 149.19.108.0/23 *Exit Node*	... ...

PATH-3 presents the communication link of Starlink users under the same GS. *PATH-4* presents the link under different GS but the same PoP. *PATH-5* presents the link different PoP.

network. The result disclosed that there were changes in the Starlink backbone routing strategies, which could also impact the network latency.

Using *StarTrace*, we obtained different routing paths including communications of Starlink users under the same GS (*Path-3*), users under different GS but the same PoP (*Path-4*), and users under different PoP (*Path-5*). The statistics of these routing paths are illustrated in Table IV. The path-tracing results indicate that Starlink supports a variety of routing strategies for internal communication. Each type of routing strategy exhibits distinct routing paths. We presented a simplified diagram with the discovered route paths in Fig. 5. Specifically, communications between users under the same GS typically complete at the *gateway node*, without the routing of the upstream network. Conversely, communications between users at different GS depend on network transit through the *link nodes* and *exit nodes*. Furthermore, we observed there were logical connections between PoP *exit nodes* of different areas, enabling user communication under different PoP without Internet transit. In other words, Starlink has constructed an internal backbone network, connecting the PoP *exit nodes* across the world, which may enhance data transmission efficiency. It also clarifies the reason why *exit nodes* worldwide are assigned IP addresses of the same subnets, *206.224.64.0/19 and 149.19.108.0/23*. We also identified the MPLS method deployed within the backbone through ICMP trace [44]. It is noteworthy that Pan et al. [7] have consistently measured the backbone topology through the recruitment of volunteers.

Based on the architecture of the Starlink backbone network, we inferred that several specific routing strategies were designed to optimize the communication from Internet users to Starlink users. The inference was validated through the following case study. In our analysis of routing paths from our cloud server in Berlin to the active Starlink hosts in Seattle. We found the probe packets first reached the PoP *exit node* in Frankfurt during April and May 2024. The Starlink backbone then routed the packets to Seattle via the PoP in Chicago (Probe→Starlink backbone→Target). However, after June 2024, the routing paths changed back to the traditional strategy, which means the probe packets reached the POP in Seattle directly through Internet transit (Probe→Internet→Target). The average latency

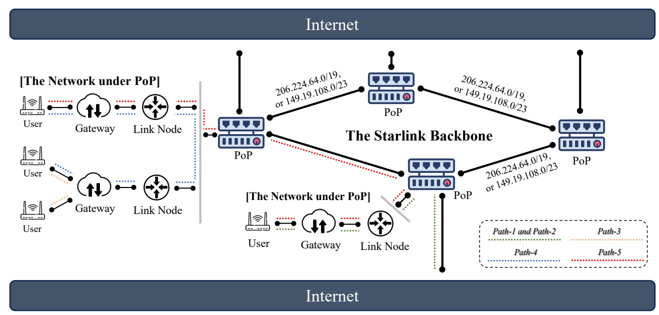


Fig. 5. The topology diagram of the whole SLTN and the multiple route paths that Starlink supports to facilitate diverse communication needs. (Path-1 and Path-2: Internet Access Service, Path-3: communication of users under the same GS, Path-4: users under different GS but the same PoP, Path-5: users under different PoP).

of the optimized routing strategy (135.20 ms) is 79.43% of the traditional (170.21 ms). We also observed a similar optimization in the path tracing from Berlin to Sydney. In summary, we found potential changes in the routing strategies of the Starlink backbone, which had impacted the latency.

Answer to RQ4: Starlink has developed multiple internal routing strategies and a specific backbone network to facilitate direct user-to-user interactions without traditional Internet transit. Each strategy employed a distinct routing path to meet diverse communication requirements. We found there were changes in routing strategies, which affected the network latency.

V. APPLICATION AND FUTURE WORK

Our measurement profiled a closed-source network system for researchers. This work is non-trivial and has the following practical applications. It also gives inspiration to future measurement studies of the SLTN.

For Starlink users, our measurement helps to improve users' perceptions of the current network status. We noticed several users complained that the network performance varied each time they used Starlink, which may be attributed to being assigned a poor network link temporarily. Through the path tracing method of *PATH-1* and the IP geolocation of HOP

3, the location of the connected PoP will be easily identified from the user's view. It also acts as an indication of ISL if the connected PoP is far from the user. These measures assist users in reporting and troubleshooting network issues. We are working on a user-friendly tool to display the current network link and network status.

For service discovery and security risk evaluation of the SLTN, we provide reliable base data including the standardized NADB and the algorithm for classifying network assets. The data revisions mitigated the errors of active services due to irrelevant scanning targets as input [9]. With the exact usage of the network assets, network service discovery engines can avoid the large-scale scanning of critical network nodes. Furthermore, our study also develops network fingerprints for essential network routing nodes, including IP address, active service, IP geolocation, and the specific role within the network. These fingerprints will help researchers identify and track the security risks within the system. For instance, we found the open BGP and router management services exposed to the Internet, which had been protected.

For QoS evaluation, previous studies ignored the effect of the Starlink terrestrial segment on the whole network. However, we found there were periodic switches of PoP that Starlink users connected and periodic changes in routing strategies of the SLTN. These two factors were both observed to cause significant variations in network latency. We believe it is highly necessary to track and monitor the network routing path while recording the performance, which helps to assess how other factors like weather and satellite movement affect QoS more accurately.

Furthermore, we propose several perspectives for future measurement. In topology recovery of the Starlink backbone, studies still highly rely on the user volunteers as probes [7]. As detailed in §IV-D, we observed that, due to the particular BGP policy, part of the routing paths from Internet servers to the Starlink users traversed the Starlink backbone network. It means that regular Internet probes are also feasible for recovering the topology with a large-scale external measurement. Besides, the exact countries and areas that each PoP covers should be measured and analyzed in depth, which helps us understand the routing strategies of the whole network. In addition, the frequency and triggers of the changes in SLTN routing strategies have also not been carefully investigated.

VI. Ethics

Our path-tracing and host-scanning strictly relied on and followed the guidelines outlined in the Menlo and Belmont reports [37]. The sending rate of probe packets in our measurement was controlled, which was suggested by previous similar measurement work [41] and had little resource consumption. Besides, our research is strictly based on data collection and analysis. The OSINT we used does not include existing academic publications or technical reports, and we have clarified potential overlaps and their differences. In addition, we carefully examined whether the OSINT data collected from the Internet was released voluntarily by individuals and organizations. The analysis of users' previous network status was of a large user base statistically, with no focus on individuals. The user's privacy was not compromised.

VII. Related Work

In this section, we provide an overview of the previous research related to the SLTN.

Architecture Measurement. During the same period of our study, Pan et al. [7] provided an empirical explanation of how Starlink works, including Starlink access networks, gateway, PoP structures, etc. However, the study lacks a systematic and specific measurement methodology and robust data analysis, which severely undermines the accuracy of the results. The explanation is also limited to the users' views, based on the observations of Starlink user volunteers. In our study, we proposed a set of specialized methods and conducted a systematic measurement of the SLTN. We not only profiled this closed network system from various aspects, but also developed a standardized dataset and effective classification algorithms of network assets based on real network status and topology. We also highlight the practical applications of our findings in future studies on security, QoS evaluations, etc.

Security Evaluation. In service discovery and security risk evaluation, Zhang et al. [8] proposed an efficient method for detecting various types of servers and their open services within the Starlink network. ZoomEye [9] performed an active host scan on some IP addresses associated with Starlink and gave statistics on the open services. Kareem et al. [18] analyzed the cybersecurity landscape surrounding Starlink, with a focus on identifying potential threats, assessing associated risks, and proposing mitigation strategies to bolster the resilience of the network. Tieby et al. [19] conducted an empirical study, revealing illicit scanning events, Mirai-based infections, source address spoofing, etc. Nonetheless, these works frequently scanned numerous network resources irrelevant to the SLTN, leading to erroneous service listings. Little attention is paid to constructing a reliable and comprehensive network assets dataset.

QoS Measurement. In QoS measurement and evaluation, a considerable amount of research has focused on evaluating the performance of the Starlink network from multiple dimensions, such as bandwidth, round-trip time, and one-way delay [12] [13] [15] [16]. These studies have also identified characteristics such as the internal adjustment every 15 seconds [14]. Traditional research is conducted with specialized hardware or recruiting large numbers of satellite Internet customers. In comparison, Izhikevich et al. [12] proposed HitchHiking, a methodology that provides global visibility into LEO. HitchHiking is the first job mainly based on Internet-exposed services that use LEO Internet and completed the most extensive study to date of Starlink network latency. Furthermore, studies have investigated how spatial factors, including weather and satellite coverage, affect the QoS [11] [17]. We note that these studies did not typically take the network status of the SLTN as a factor affecting QoS, whereas our findings revealed that the status was not stable.

VIII. CONCLUSION

Starlink has emerged as a preeminent satellite Internet service provider with LEO satellites. We noticed the terrestrial segment, as an essential component of the Starlink network, gained less attention from researchers. In this paper, we conducted a thorough measurement study to profile the SLTN and reveal its potential effects. Since the SLTN is a closed-source and widely distributed system with network fluctuations, a specific measurement framework was proposed to collect various research data. A set of novel findings was then distilled through in-depth data analysis. We presented the scale, distribution, and features of the network assets, mapped the consistent topology, and identified four types of routing nodes. An effective algorithm was proposed to classify the network assets. We also uncovered multiple internal routing strategies. Particularly, we emphasized and demonstrated the practical applications of our measurement. We observed the switches of terrestrial infrastructure and the changes in routing strategies, which should be considered in QoS evaluations. It also provides insights for improving users' perceptions and studies like security risk evaluation. Moving forward, we believe it is necessary to continuously monitor the SLTN architecture and its behavior. Further research on Starlink needs to get a more comprehensive view of the terrestrial segment and carefully consider its effect.

ACKNOWLEDGMENT

We sincerely thank all potential "volunteers" in Starlink communities who inadvertently shared their network status on social media to help improve the QoS. We are grateful to the anonymous reviewers for their insightful comments. This work is supported by the project (No. E3H02611E1) of the National Key Laboratory of Science and Technology on Blind Signal Processing (No. 23007522).

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